1. PRD — Product Requirements Document

Project: Advanced Personal Portfolio Website with AI Video Intro + Advanced Admin Panel **Builder:** Google AI Studio (Build / Max mode) **Source spec:** *Advanced_Portfolio_Website_Feature_Spec_Roman_Urdu.md* **Version:** 1.0

> **Roman Urdu summary:** Yeh document batata hai ke website *kya* karegi aur *kyun*. Har feature ka ID hai (FR-xx public site, AD-xx admin), taake AI builder ya aap kisi bhi feature ko tracking list mein check kar sakein. Pehle dummy (sample) data se website poori chalay gi; baad mein admin panel se aap real data daal kar dummy data replace kar denge.

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1. Product Overview

A premium, dark, cinematic personal portfolio. On a first visit the visitor sees a full-screen **AI avatar introduction video**; when it ends (or is skipped) the site is revealed through a smooth cinematic transition into the hero. Behind the public site sits a secure, **advanced admin panel** so the owner can manage every piece of content (projects, skills, services, intro video, SEO, inquiries, settings) without touching code.

Two products in one codebase

1. **Public site** — for clients, recruiters, collaborators.
2. **Admin panel (`/admin`)** — for the owner (and optional editors).

2. Goals and Success Metrics

Goal	Metric / Target
Visitor understands who the owner is in seconds	Hero shows name, role, value proposition above the fold on every device
Memorable first impression	Intro video plays with working Skip / Replay / captions; never blocks access
Convert visitors to enquiries	Contact form works end-to-end; inquiry conversion tracked in admin analytics

Goal	Metric / Target
Fast and stable	Mobile Lighthouse: Performance ⩾ 85, Accessibility ⩾ 95, SEO ⩾ 95, Best Practices ⩾ 95; LCP < 2.5 s, CLS < 0.1, INP < 200 ms
Owner independence	100 % of visible content editable from admin; no code edits to update content
Trustworthy	No fake testimonials, stats, logos or links in the **published** state (see §5)

3. Audience and Personas

- **Client Clara** — small-business owner hiring a freelancer. Wants proof of results, services, price clarity, easy contact (form/WhatsApp).
- **Recruiter Rafay** — screens many candidates quickly. Wants skills, experience timeline, downloadable CV, clear project roles.
- **Collaborator Zain** — creative peer. Wants to see process, case studies, personality, social links.
- **Owner (Admin)** — wants fast content updates from phone or laptop, clear inbox of inquiries, simple analytics, zero risk of breaking the site.

4. Scope

4.1 Must have (launch / MVP)

Intro video + cinematic reveal, sticky navigation, hero, about, skills/tools, experience/resume, projects gallery (filters, search, details), services, contact system, dark/light theme, SEO basics, accessibility, responsive design, admin panel (all modules in §8), sample-data system.

4.2 Should have (included, behind feature flags)

Testimonials, FAQ, pricing packages, achievements/certifications, command palette (Ctrl/Cmd+K), blog/insights, language switcher, first-party privacy-friendly analytics, social share.

4.3 Could have (second stage, flagged OFF by default)

Heavy 3D hero scene, custom cursor, magnetic buttons, RSS feed, booking-calendar embed, newsletter.

4.4 Out of scope

Visitor accounts/login, e-commerce/payments, comments system, live chat, multi-tenant (multiple portfolios).

5. Content Principles and the Dummy-Data Policy

The spec forbids fabricated testimonials, statistics and clients. The owner also wants a fully populated site before real data exists. These are reconciled as follows:

1. **Every content row has an `is_sample` flag.** The seed (dummy) data is created with `is_sample = true`.
2. Sample content is **clearly obvious in admin**: yellow “SAMPLE” badge on each row, a dashboard widget “Sample content remaining: N items”.
3. On the public site, sample content is shown normally while the site is in **Preview/Demo** state, but a **Launch Readiness check** in admin warns (and requires explicit confirmation) before publishing while sample items remain.
4. Sample text is realistic but generic (e.g. “Sample Project One”), uses placeholder images generated locally (gradients/SVG, no third-party logos or celebrity photos), and **never uses real company names, real people or real quotes**.
5. Admin has a **“Replace sample data” wizard** and a **“Delete all sample data”** button (see AD-30).
6. Public site shows buttons such as Live Demo / Source / CV Download **only when a real value exists**; sample records point to no external URL.

6. Functional Requirements — Public Site

6.1 Intro video and cinematic reveal

- **FR-01** First eligible visit shows a full-viewport intro screen with poster image instantly visible and video loading state.
- **FR-02** Video autoplays **muted**; sound only after explicit user click/tap on an Unmute control.

- **FR-03** Controls: Skip Intro (always visible and functional), Play/Pause, Mute/Unmute, Captions toggle (WebVTT).
- **FR-04** The transition starts on the video's real `ended` event or on Skip; never relies on a timer alone. Transition runs **once** (guarded).
- **FR-05** Transition: final frame scales/fades/masks into the hero, with subtle light sweep; no white flash, blank screen or layout shift. Portfolio is already rendered underneath.
- **FR-06** Reveal sequence: navbar → hero heading (word/line) → intro text → CTA buttons (staggered).
- **FR-07** Preference stored in `localStorage` (try/catch): "skip intro on future visits". Does not replay on internal navigation. **Replay Intro** link in footer/menu.
- **FR-08** Fallbacks: video error, slow load (timeout), unsupported format, autoplay rejected, `prefers-reduced-motion`, storage failure → site opens immediately or shows a simple tap-to-play; **site access is never blocked**.
- **FR-09** Intro can be turned off globally from admin (Intro manager) and desktop/mobile video sources can differ.

6.2 Navigation

- **FR-10** Sticky glass navbar: logo/name, Home, About, Projects, Services, Resume, Contact, "Let's Work Together" CTA.
- **FR-11** Active-section highlighting (scroll-spy) on Home; route highlighting elsewhere; smooth anchor scroll with offset for sticky bar.
- **FR-12** Compact state on scroll; optional hide-on-scroll-down/show-on-scroll-up; scroll progress line.
- **FR-13** Mobile hamburger: tap, keyboard, Escape and outside-tap close; focus trap while open; closes after link click.
- **FR-14** Nav items, order and CTA label are driven from admin settings.

6.3 Hero

- **FR-20** Name, professional title, 1–2 line value proposition, portrait or abstract visual, "Explore My Work" and "Contact Me" CTAs, real social links, scroll indicator, optional availability badge ("Available for projects").
- **FR-21** Subtle heading reveal; optional slow role rotation (pausable, not rapidly changing); soft gradient/orb background.
- **FR-22** Every CTA scrolls/navigates to its correct destination.

6.3b Global animation system

- **FR-25** Reusable motion tokens: scroll reveal, stagger, hover lift, image zoom, page transition, scroll progress, optional parallax, optional magnetic buttons/custom cursor (off on touch).
- **FR-26** Never hijack scroll; respect `prefers-reduced-motion`; content visible if animation JS fails; no unbounded infinite loops.

6.4 About, skills, experience

- **FR-30** Short + long biography, portrait, values/philosophy, call-to-action.
- **FR-31** Skills grouped by category (Design, Development, AI tools, Content/Video, Marketing/Business, Other) — **no fake percentage bars**; each skill may link to related projects.
- **FR-32** Tools wall with accessible labels.
- **FR-33** Experience, education and certification timeline with staggered reveal.
- **FR-34** Achievements list (verified only; counters only for admin-entered real numbers).
- **FR-35** `/resume` page: structured CV + **Download CV (PDF)** button shown only when a PDF is uploaded; print-friendly stylesheet.

6.5 Projects

- **FR-40** Home shows featured projects (large featured card + grid). `/projects` shows full gallery.
- **FR-41** Category filter chips, live search (title, description, tech), sort (Featured / Newest / Category), pagination or Load-more, empty-results state, lazy-loaded images with alt text.
- **FR-42** Card interactions work on hover, keyboard focus and touch (no hover-only information).
- **FR-43** `/projects/:slug` case-study template: summary, cover/demo video, problem, approach, role, process, gallery (accessible lightbox), tools, challenges & solutions, results (only if entered), Live Demo / Source buttons (only if URLs exist), related projects, previous/next, CTA “Discuss a similar project”.
- **FR-44** Stable shareable URLs, Open Graph preview per project, proper 404 for invalid slug, browser back/forward works.
- **FR-45** Native Share API with copy-link fallback.

6.6 Services, trust, FAQ

- **FR-50** Service cards: problem solved, deliverables, process, related projects, optional starting price / “Request a Quote”, CTA “Discuss this service” (preselects service in contact form).
- **FR-51** Optional pricing packages, FAQ accordion, booking link.
- **FR-52** Testimonials: only approved entries; manual prev/next; autoplay (if enabled) has pause control and honours reduced motion. Section auto-hides when no approved testimonials exist; “My Process” section shown instead.

6.7 Contact

- **FR-60** Form: name, email, service/project type, optional budget and timeline, message, consent checkbox (if privacy notice enabled).
- **FR-61** Client + server validation, inline errors, loading state, success shown **only after a real successful server response**, failure with retry.
- **FR-62** Anti-spam: honeypot field, rate limit, bot-challenge token (Cloudflare Turnstile or equivalent), server-side verification.
- **FR-63** Email/WhatsApp/social links from admin settings; copy-email button with confirmation; optional booking link.
- **FR-64** Each inquiry is stored and visible in admin inbox; owner receives email notification through a configured provider.

6.8 Personalization and extras

- **FR-70** Dark/light/system theme with saved preference and no flash of wrong theme.
- **FR-71** Command palette (Ctrl/Cmd+K): navigate sections, search projects, toggle theme, replay intro; Escape closes; focus managed.
- **FR-72** Language switcher (English / Roman Urdu / Urdu RTL) shown **only if** a language has complete translations in admin.
- **FR-73** Blog/Insights (flag): listing, filters, article page, reading time, related posts, SEO.
- **FR-74** Privacy policy page (editable in admin), custom 404 page, cookie/analytics notice only if tracking requires it.
- **FR-75** First-party, privacy-friendly analytics (page views, project views, CTA clicks, form conversions); no form content collected.

6.9 SEO and sharing

- **FR-80** Unique title and meta description per page (editable in admin), semantic headings, canonical URL, Open Graph/Twitter cards, `sitemap.xml`, `robots.txt`, JSON-LD (`Person`, `WebSite`, `CreativeWork`) with accurate fields only; admin and staging routes `noindex`.

7. Non-Functional Requirements

- **Performance:** image formats WebP/AVIF, intro video H.264 MP4 (WebM optional) with poster, lazy-load below-fold media, route-level code splitting, reserved media dimensions, 3D/heavy effects loaded on demand, simplified effects on low-power devices.
- **Accessibility (WCAG 2.2 AA):** semantic HTML, keyboard operability, visible focus, contrast ≥ 4.5:1 text, labelled inputs and error association, captions on intro, reduced motion support, focus management in dialogs/menus.
- **Responsive:** mobile portrait/landscape, tablet, laptop, wide desktop; no overflow or overlap; face never cropped in intro video.
- **Security:** secrets never in client bundle; server-side authorization; input validation and sanitization; rate limiting; safe external links (`rel="noopener noreferrer"`).
- **Privacy:** localStorage only for non-sensitive preferences (theme, intro skip, language); minimal personal data in inquiries; data-retention setting in admin.
- **Reliability:** every external asset failure degrades gracefully; error boundaries; friendly error pages.
- **Browser support:** last 2 versions of Chrome, Edge, Firefox, Safari (iOS 15+), Samsung Internet.

8. Functional Requirements — Admin Panel (Advanced)

Admin lives at `/admin`, is `noindex`, is protected by **server-enforced authorization** (database Row Level Security), not just hidden routes.

8.1 Access and security

- **AD-01** Email + password login via managed auth provider; optional TOTP two-factor; password reset; session timeout; logout everywhere.

- **AD-02** Roles: **Owner** (everything), **Editor** (content only, no users/settings/danger zone). Owner can invite/remove editors.
- **AD-03** Login rate limiting and generic error messages; audit log records sign-ins and every content change (who, what, when).

8.2 Dashboard

- **AD-10** KPI cards: views (7/30 days), top projects, new inquiries, conversion rate.
- **AD-11 Launch Readiness checklist:** profile complete, intro video uploaded, CV uploaded, ≥ 3 real projects, contact email set, SEO set, sample content count — each item links to the screen that fixes it.
- **AD-12** Recent inquiries, recent activity, quick actions (New project, Upload media, View site, Replay intro).

8.3 Content modules (all with create / edit / duplicate / delete / reorder / draft / publish / schedule / preview)

- **AD-20 Projects:** rich-text case study editor, cover + gallery upload with drag-and-drop ordering, alt-text required, categories, tags, tools, role, dates, links (validated), featured flag, SEO fields, slug auto-generation with uniqueness check, revision history with restore.
- **AD-21** Categories and tags manager.
- **AD-22** Skills & tools manager (groups, icons, linked projects).
- **AD-23** Experience, education, certifications, achievements.
- **AD-24** Services, packages, FAQs.
- **AD-25** Testimonials (pending / approved / hidden, permission-confirmed checkbox, photo optional).
- **AD-26** Blog posts (flag-controlled) with same editor.
- **AD-27** Legal pages (privacy) editor.

8.4 Site control

- **AD-30 Sample Data Manager:** list of all sample items by module; per-item “Replace with real” shortcut; **Delete all sample data** (type-to-confirm); counter on dashboard.
- **AD-31 Intro Manager:** upload/replace desktop and mobile video, poster, captions (.vtt), scripts (English/Roman Urdu), enable/disable, autoplay-muted option, timeout seconds, “skip preference” default, live preview + test-fallback buttons.

- **AD-32 Profile & Site Settings:** name, title, short/long bio, portrait, availability text, contact email, WhatsApp, social links, booking URL, accent color from approved palette, default theme.
- **AD-33 Sections & Navigation:** toggle visibility and order of homepage sections, edit nav items and CTA label, feature flags (blog, testimonials, command palette, language switcher, analytics, 3D, custom cursor).
- **AD-34 SEO manager:** global defaults and per-page title/description/OG image, sitemap preview, indexing toggle for staging.
- **AD-35 CV manager:** upload/replace/remove PDF.
- **AD-36 Media Library:** upload (image/video/pdf/vtt), automatic size limits and compression guidance, alt-text editing, search, “where used”, replace file keeping URL, delete protection when in use.

8.5 Inbox and analytics

- **AD-40 Inquiries inbox:** list, search, filter by status (New, In progress, Replied, Closed, Spam), assign status, private notes, reply via mailto/WhatsApp link, mark spam, export CSV, bulk actions, auto-delete after retention period.
- **AD-41 Analytics:** page views, unique sessions (anonymous), top projects, referrers, CTA clicks, form starts vs submits; date range filter; export CSV.

8.6 Data safety

- **AD-50** Export full content as JSON and import (validated, preview-diff before applying).
- **AD-51** Audit log viewer (filterable). Revision history for projects/posts/pages.
- **AD-52** Danger zone (Owner only): delete all inquiries, reset to sample data, with type-to-confirm.
- **AD-53** Admin is fully responsive and usable on mobile.

9. Feature Flags (editable in admin)

`intro_enabled`, `testimonials_enabled`, `faq_enabled`, `pricing_enabled`, `blog_enabled`, `command_palette_enabled`, `language_switcher_enabled`, `analytics_enabled`, `three_d_hero_enabled`, `custom_cursor_enabled`, `scroll_progress_enabled`, `hide_on_scroll_nav`, `sample_banner_enabled`.

10. Dependencies and Assets the Owner Must Provide

Essential: display name, professional title, bio, AI intro video (+ poster, optional mobile crop, captions), portrait, 3–6 real projects (images, descriptions), real skills/tools, services, contact email, social URLs, CV PDF. **Optional:** approved testimonials, certifications/awards, pricing, WhatsApp/booking URL, blog content, translations. **Accounts/config:** database + auth + storage project, email provider API key, bot-challenge keys, hosting account, custom domain.

11. Risks and Mitigations

Risk	Mitigation
Autoplay blocked by browser	Muted autoplay; fallback to poster + big Play button
Large video slows site	Compressed MP4, poster first, lazy preload strategy, separate mobile file
SPA hurts SEO	Pre-render public routes at build time; per-route meta (see TRD §6)
Admin misconfiguration exposes data	RLS enabled on every table, tested policies, no service key in client
Sample content accidentally published	Launch Readiness warning + Sample Data Manager
Over-animation hurts usability	Motion tokens, reduced-motion mode, performance budget

12. Definition of Done

A feature is done when: it works on mobile and desktop; keyboard accessible; has loading/empty/error states; has seed data; is editable in admin where applicable; passes the checklist in the Implementation Plan; and no placeholder button or console error remains. If any provider key is missing, the UI states exactly what must be configured instead of pretending to work.

13. Open Decisions for the Owner

1. Final accent color (default: electric violet; alternative: electric blue).
2. Languages to support at launch (default: English only; Roman Urdu/Urdu if translations supplied).
3. Whether the blog and testimonials are enabled at launch.
4. Custom domain and hosting provider.